

Once the hardware design on the HDL level is complete, schematic designs can be used to visually present the circuit design on a computer. A Schematic editor not only presents a graphical view for placing components, they can also be used to verify the function of circuits by simulating the circuit and running tests on them.

Once logical and schematic designs are completed, the SoC needs to be fabricated. For that, the circuit designs need to be converted into physical interconnection on a silicon die. Since modern chips have billions of transistors interconnected, manual placement of each transistor on a circuit is impractical. Layout tools take care of that. These tools are used to automate the physical design of a circuit. The complete IC can be large and may contain redundant circuits placed at different positions (e.g. Each processing core is going to need its own set of adders). Layout tools take modular circuits and use them to build larger circuits which can be verified and tested on a simulator.

Simulation and Testing

Although we are presenting it separately, testing of design at various stages go alongside the designing process. A logical problem on the HDL level would mean that both schematic design and layout of the chip would change. For each level of design, the testing and verification is done using tools designed for that level of design. For example: a HDL testing tool would be able to verify if the circuit defined using HDL can perform the desired



Schematic Design of a Circuit